$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} = 0$$

$$-(\lambda - 1)^2 - 1 = 0$$

$$-(1 - \lambda^2) - 1 = 0$$

$$\lambda = \pm\sqrt{2}$$

$\Rightarrow$ Eigenvalues of A^{13} is $\lambda^{13} = \pm 64\sqrt{2}$

$$Ax = \lambda x$$

$$A^2 x = A(Ax) = A\lambda x = \lambda(Ax) = \lambda^2 x$$

$$\vdots$$

Generalising for $A^n x = \lambda^n x$